

Randomized and low-rank approximation

Optimal matvec query complexity of HODLR approximation

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging because matching adaptive-query lower and upper bounds must cover depth and accuracy jointly; community impact is the cost of hierarchical matrix compression.

Area: information complexity of hierarchical matrix approximation

Context and notation

Use exact real arithmetic, with comparisons, standard Gaussian sampling, and exact SVDs available as primitives; charge an SVD of an $a \times b$ matrix $O(ab \min(a, b))$ operations. This states the idealized arithmetic model used here, rather than a finite precision or bit complexity claim. A matrix–vector query returns either Av or $A^\top v$ for one chosen real vector v ; both types count toward the total. Randomized guarantees are for every fixed input, with probability or expectation over the algorithm’s randomness.

Problem statement

Let $n = 2^L k$ with integers $k \geq 1, L \geq 2$, and $0 < \varepsilon < 1/2$. Define $\mathcal{H}_{n,k}$ recursively: a matrix of order at most k is unrestricted; at a larger node, the two equally sized diagonal blocks must satisfy the same definition, and each off-diagonal block must have rank at most k .

Let $q_*(n, k, \varepsilon)$ be the smallest worst-case number of queries with which a randomized adaptive algorithm, given only oracle access to an arbitrary $A \in \mathbb{R}^{n \times n}$, returns $B \in \mathcal{H}_{n,k}$ such that

$$\Pr \left[\|A - B\|_F \leq (1 + \varepsilon) \min_{H \in \mathcal{H}_{n,k}} \|A - H\|_F \right] \geq 0.99$$

for every A . Here only oracle calls are charged: arbitrary measurable computations between queries are allowed. This is an information complexity question.

Question

Determine $q_*(n, k, \varepsilon)$ up to universal multiplicative constants, including its joint dependence on k, L, ε .

Reference

Tyler Chen, Feyza Duman Keles, Diana Halikias, Cameron Musco, Christopher Musco, and David Persson, *Near-optimal hierarchical matrix approximation from matrix-vector products*, SODA 2025, 2656–2692, Definition 1.1, Problem 1.1, Theorems 1.1–1.2, end of §6, and §8.

The known upper bound is $O(\min\{n, kL^4/\varepsilon^3\})$. The source proves $\Omega(kL + k/\varepsilon)$ in its regime $n \geq c_0 k/\varepsilon$ for a sufficiently large absolute c_0 ; this restriction matters near the full-recovery threshold.

Status evidence

The latest arXiv version is v2, October 24, 2024. Searches for “HODLR query complexity 2026” and “HODLR approximation 2026 lower bound” found no matching improvement. Musco’s [February 2026 ICERM slides](#), numbered slide 14, retain the stated upper bound. No later explicit open-status confirmation was located.

Audit — 2026-09-10

Rechecked [Chen et al., Theorems 1.1–1.2 and §8](#). The gap in logarithmic and accuracy factors remains. HODLR query-complexity and later hierarchical-approximation searches found no matching characterization, including the near-full-recovery regime.